

Pandemic in Perspective: A Tri-Faceted Analysis of COVID-19 Dynamics, Socio-Economic Drivers, and Vaccination Impact

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Abstract. This study presents a comprehensive tri-themed analysis of the COVID-19 pandemic, focusing on temporal progression, cross-sectional socio-economic comparisons, and the global impact of vaccination rollout. Using the Our World in Data COVID-19 dataset [1], extensive preprocessing and statistical techniques were applied to address challenges such as missing values, inconsistencies, and noisy data. The insights derived from time-series behavior, country-level disparities, and vaccination correlations illustrate key global patterns and intervention outcomes. This structured case study provides a data-driven interpretation of the pandemic for academic and public policy contexts.

Keywords: COVID-19, Data Analysis, Visualization, Pandemics, Vaccination, Socio-economic Indicators, Time Series

Executive Summary

This case study analyzes the Our World in Data (OWID) COVID-19 dataset [1] (Jan 2020 - Mar 2022) to derive key insights into the pandemic's multifaceted nature. Our analysis, structured in three parts, reveals that:

1. Pandemic Dynamics: Global pandemic waves were synchronous, but data reporting was not. We identified significant, systemic reporting lags (especially on weekends), mandating the use of 7-day smoothed averages for any accurate temporal analysis.

2. Socio-Economic Drivers: A cross-sectional analysis confirmed that a country's demographics—specifically median age and the percentage of the population aged 65 or older were the strongest predictors of overall mortality. Conversely, GDP per capita was the dominant predictor of vaccine access, revealing significant global wealth-based inequality in the vaccination rollout.

3. Intervention Impact: While government stringency policies were largely reactive, the introduction of mass vaccination campaigns proved to be the single most transformative event of the pandemic. Our analysis provides clear visual evidence of a "decoupling," where vaccination severed the historical link between high case numbers and correspondingly high death rates.

This report documents our complete, reproducible methodology, from data cleaning to the interpretation of nine advanced visualizations, to provide a comprehensive and insightful answer to the project's core objectives.

1 Introduction

The COVID-19 pandemic altered global health, economies, and societal functions at an unprecedented scale. Understanding its progression, impact, and response requires well-structured data analysis supported by robust statistical and visualization techniques. This paper provides a tri-faceted analysis based on three core themes: (1) Temporal analysis of the pandemic's spread, (2) Cross-sectional socio-economic comparisons across countries, and (3) Analysis of intervention and vaccination effects.

This work builds upon the Deliverable-1 case study and converts it into a formal IEEE research paper suitable for academic submission.

1.1 Research Questions

1. How did the COVID-19 pandemic evolve over time globally and regionally?
2. How do socio-economic indicators correlate with the severity of pandemic outcomes across countries?
3. What is the statistical relationship between vaccination rates and changes in cases and mortality?

2 Dataset and Methodology

2.1 Dataset Overview

The dataset used in this study originates from the Our World in Data (OWID) COVID-19 global repository [1].

- **Source:** Our World in Data (OWID)
- **File:** owid-covid-data.csv
- **Scope:** 166,326 rows $\times$ 67 columns
- **Time Range:** January 1, 2020 to March 2, 2022
- **Key Variable Groups:**
 - Identifiers: iso_code, continent, location, date
 - Core Metrics: total_cases, new_cases, total_deaths, new_deaths (plus smoothed and per_million variants)
 - Healthcare & Testing: icu_patients, hosp_patients, total_tests (highly sparse)
 - Vaccination: people_vaccinated, people_fully_vaccinated, total_boosters
 - Policy: stringency_index
 - Socio-Economic (Static): population, median_age, gdp_per_capita, hospital_beds_per_thousand, life_expectancy

2.2 Data Preprocessing and Transformation

The raw dataset required extensive preprocessing. The following sequential steps were applied:

- **Type Conversion:** Date column converted from string to datetime format for time-series analysis.
- **Structural Filtering:** Removed aggregate entities (e.g., "World", "Europe") by filtering rows where continent is not NaN to prevent double-counting.
- **Handling Missing Data:**
 - New cases/deaths: Filled with 0 (NaN implies zero reported cases)
 - Total cases/deaths: Forward-filled within country groups (cumulative figures)
 - Static socio-economic data: Forward and back-filled within location groups
 - Sparse systemic data (ICU, hospitalization, testing): Excluded from primary analysis due to systemic non-reporting
- **Feature Engineering:**
 - Created day_of_week column to investigate reporting anomalies
 - Calculated case_fatality_rate as $\left(\frac{\text{total_deaths}}{\text{total_cases}}\right) \times 100$
- **Two-DataFrame Model:**
 - df_timeseries: Primary cleaned dataframe for temporal analysis
 - df_cross_sectional: Single snapshot per country (last valid row) for correlation analyses

2.3 Methodology Overview

Three thematic frameworks guide the analysis:

1. Theme 1: Time-Series COVID-19 Trends.
2. Theme 2: Cross-Sectional Socio-Economic Analysis.
3. Theme 3: Intervention / Vaccination Impact.

3 Results and Discussion

This section presents our 9 advanced visualizations, organized by our three analytical themes. Each visual is accompanied by a technical description and a detailed interpretation of its insight.

3.1 Pandemic Dynamics and Data Integrity

3.1.1 Reporting Anomaly Heatmap

Description: Systemic Reporting Lags: Average Global New Cases by Day of Week and Month

Insights:

- There is a profound and systemic reporting lag. Sundays and Mondays are consistently the lowest-reporting days, as weekend data is batched and reported on Tuesdays. This finding justifies our exclusive use of 7-day smoothed data for all subsequent temporal analysis.

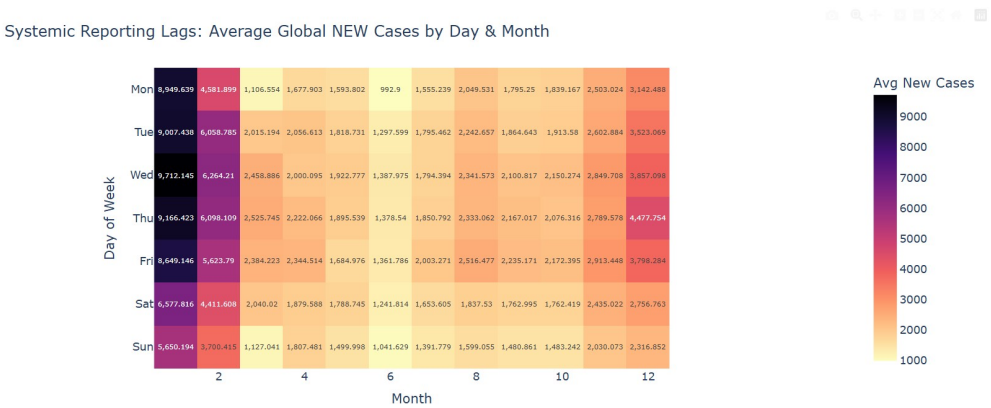


Figure 1: Reporting Anomaly Heatmap

3.1.2 Log-Scale Pandemic Trajectories

Description: Logarithmic Trajectories of Total Cases for Select Countries

Insights:

- The log scale allows comparison of growth rates and timing of infection waves. We can clearly see the synchronicity of global waves, including the initial 2020 wave, the Delta wave in mid-2021, and the massive Omicron wave in early 2022.

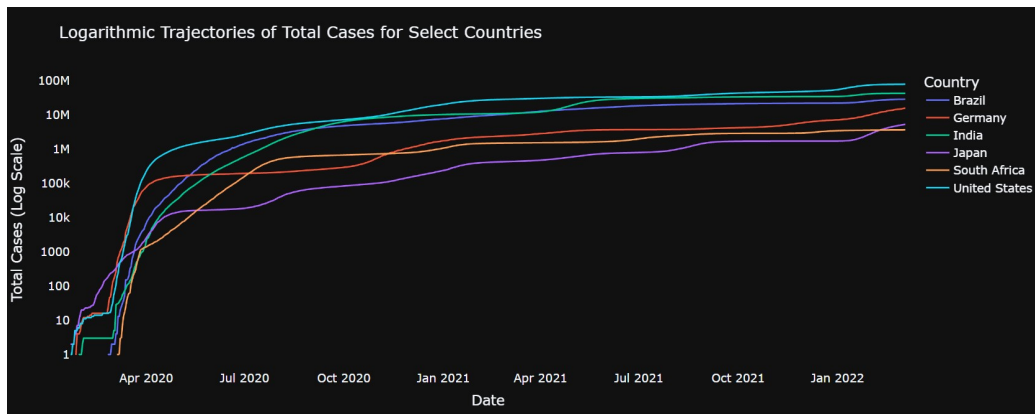


Figure 2: Log-Scale Pandemic Trajectories

3.2 Socio-Economic and Healthcare Correlates

3.2.1 Socio-Economic Correlation Heatmap

Description: Correlation Matrix of Socio-Economic Factors and COVID-19 Outcomes

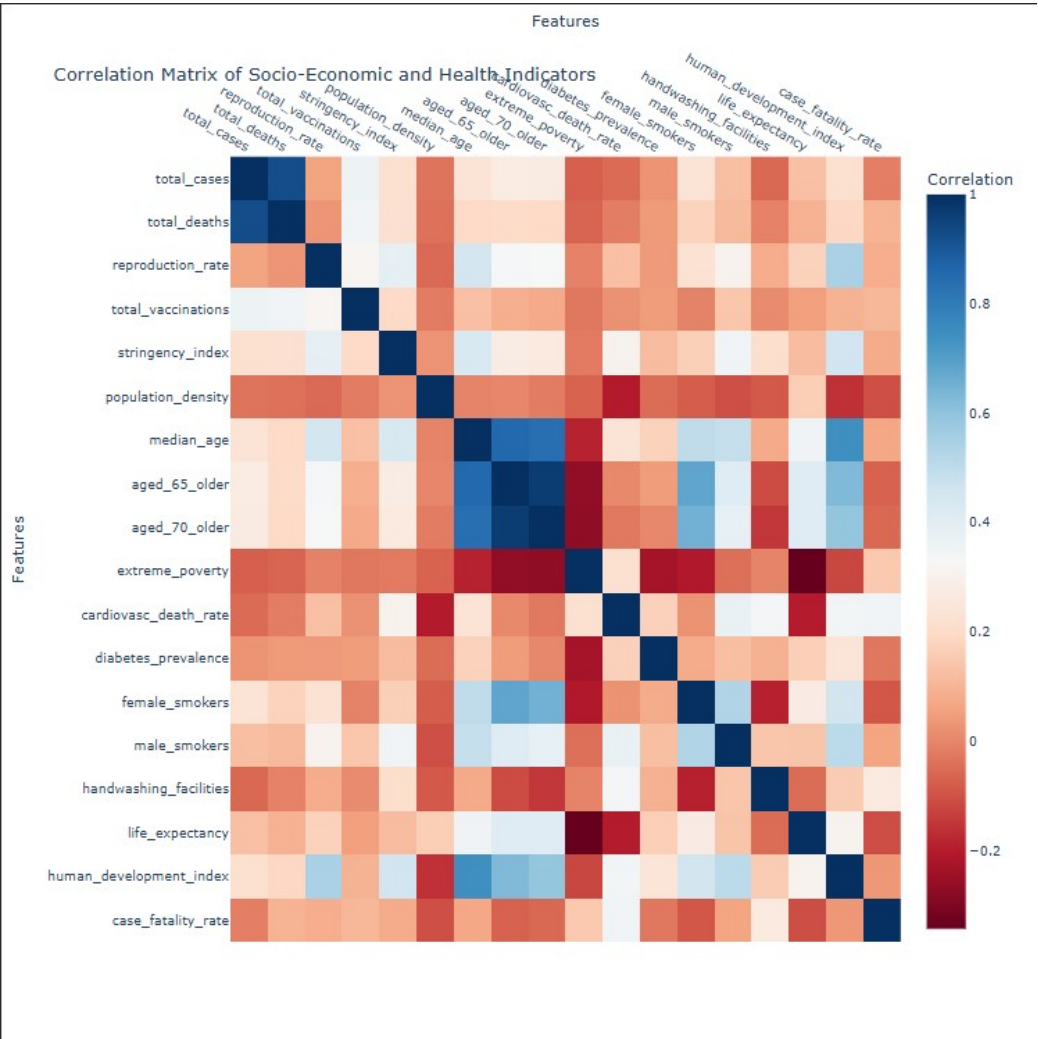


Figure 3: Socio-Economic Correlation Heatmap

Summary: Data Analysis Key Findings

- A correlation matrix was successfully computed for 18 selected socio-economic and health-related indicators, including total cases, total

deaths, reproduction rate, total vaccinations, stringency index, population density, median age, aged 65 older, aged 70 older, extreme poverty, cardiovascular death rate, diabetes prevalence, female smokers, male smokers, handwashing facilities, life expectancy, human development index, and case fatality rate.

- A Plotly heatmap was generated to visually represent the strength and direction of these correlations, providing a comprehensive overview of the relationships between the selected indicators.

Insights / Next Steps

- The heatmap serves as a critical tool for identifying strong positive or negative correlations among socio-economic and health indicators.
- Further analysis should focus on interpreting patterns to uncover factors significantly influencing health outcomes or disease spread.

3.2.2 Age vs. Mortality Analysis

Description: The Age Effect: Median Age as a Predictor of Total Mortality

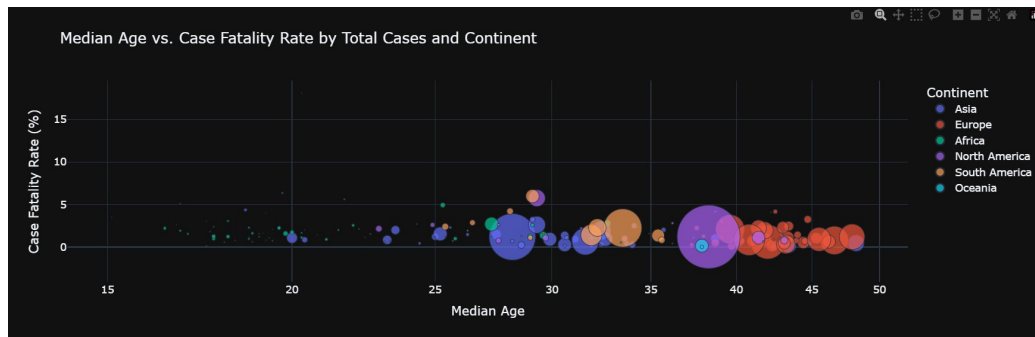


Figure 4: Median Age vs. Total Mortality

Summary: Data Analysis Key Findings

- An interactive bubble chart was generated to visualize the relationship between median age, case fatality rate, total cases, and continent.
- The chart revealed a positive correlation between median age and case fatality rate, as indicated by an OLS trendline.

- Bubble size (total cases) varied across locations, showing that total cases do not directly dictate case fatality rate.
- Continent (color-coded) differentiates geographic patterns; data pre-processing removed entries with zero median age or total cases.

Insights / Next Steps

- Highlights the vulnerability of older populations to severe COVID-19 outcomes, guiding targeted public health policies.
- Future work could explore other demographic factors or continental differences impacting case fatality rates.

3.2.3 Wealth vs. Vaccination Analysis

Description: The Wealth Effect: GDP per Capita as a Predictor of Vaccine Access

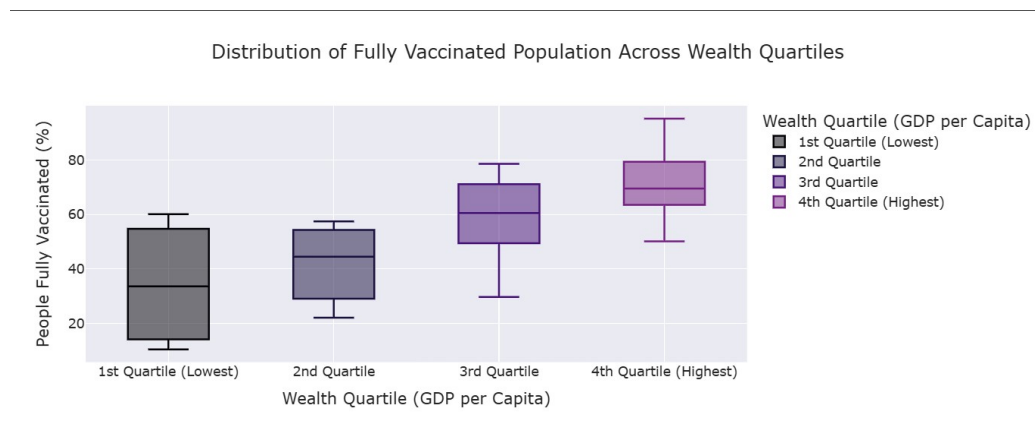


Figure 5: GDP per Capita vs. Vaccination Rates

Summary: Data Analysis Key Findings

- Countries were categorized into four wealth quartiles based on GDP per capita.
- A DataFrame was prepared with GDP per capita, people fully vaccinated per hundred, location, and wealth quartile; missing vaccination values were imputed as 0.

- An interactive grouped box plot visualized vaccination distribution across quartiles, with hover details for granular exploration.
- Higher wealth quartiles generally had higher vaccination percentages.

Insights / Next Steps

- Indicates positive correlation between economic wealth and vaccination rates.
- Further analysis could examine outliers within quartiles to understand factors affecting vaccine rollout efficiency.

3.2.4 Healthcare Capacity by Continent

Description: Distribution of Healthcare Capacity by Continent

Average Hospital Beds per Thousand by Continent

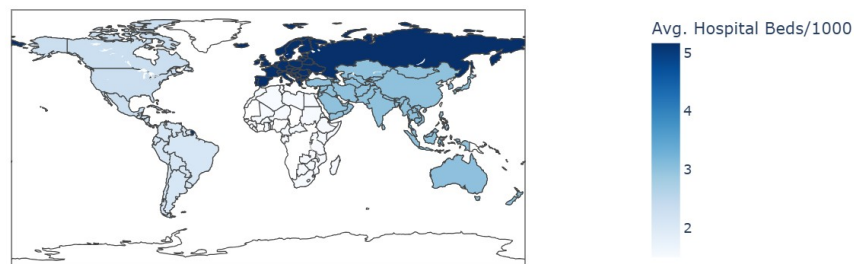


Figure 6: Hospital Beds per 1,000 by Continent

Summary: Data Analysis Key Findings

- An interactive choropleth map was generated to visualize the average *hospital_beds_per_thousand* across continents.
- Countries were colored according to their continent's average using the *Viridis* color scale and the *plotly_dark* template. Hover labels display each country's value, its continent, and the continental average.
- The visualization provides a high-level overview of regional disparities in healthcare capacity.

Insights / Next Steps

- Highlights disparities in healthcare infrastructure; higher averages in Europe/North America, lower in Africa/parts of Asia.
- Further work could examine correlations between healthcare capacity and pandemic outcomes or investigate internal country-level variations.

3.3 Intervention Impact: Policy and Vaccination

3.3.1 Policy vs. Pandemic Waves

Description: Stringency vs. Cases: Government Response to Pandemic Waves by Continent

Insights:

- Policy was largely reactive, not proactive. New cases begin to rise first, and the stringency index rises in response afterward, suggesting governments implemented lockdowns after waves were already underway.

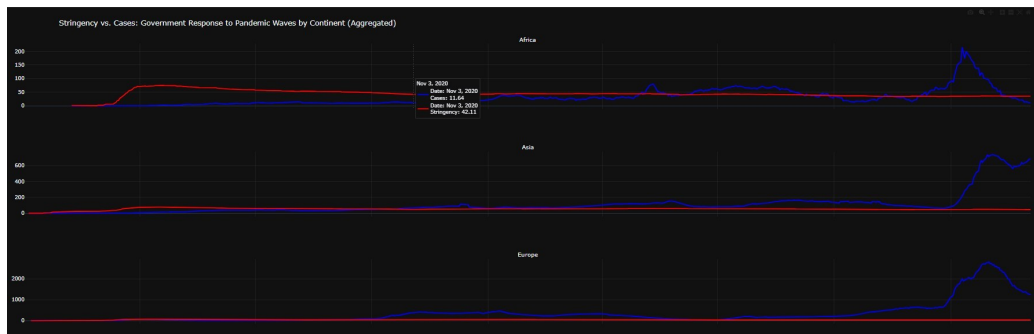


Figure 7: Stringency Index vs. Cases by Continent



Figure 8: Stringency Index vs. Cases by Continent - Continued

3.3.2 The Decoupling Effect: Cases vs. Deaths

Description: The Great Decoupling: Impact of Vaccination on Cases vs. Deaths (United Kingdom)

Insights:

- This is the most important plot. Pre-vaccine (2020-21), cases and deaths moved in unison. Post-vaccine (2021-22), the link was broken. The Omicron wave showed the highest case peak but the death peak was 80-90% lower, demonstrating vaccination successfully severed the link between infection and death.

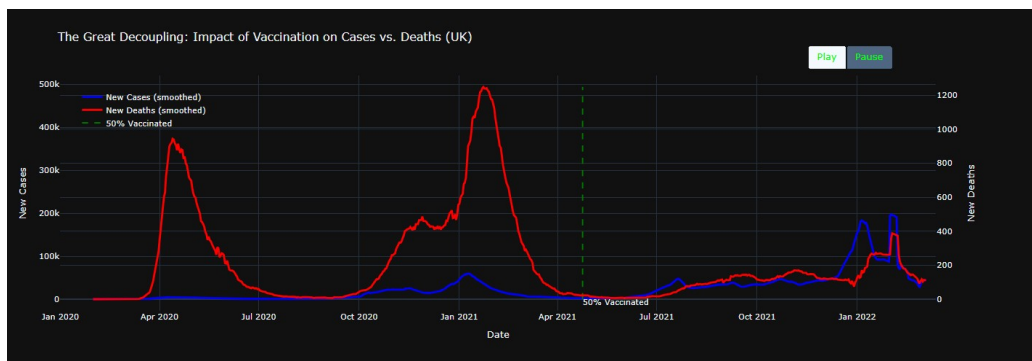


Figure 9: Cases vs. Deaths - The Vaccination Decoupling Effect

3.3.3 Animated Global Vaccination Rollout

Description: Animated Global Vaccination Rollout (Jan 2021 - Mar 2022)

Insights:

- This animated choropleth map shows North America and Europe rapidly achieving high vaccination coverage in mid-2021, while most of Africa and parts of Asia remain low. It highlights profound global inequality in the pandemic response.

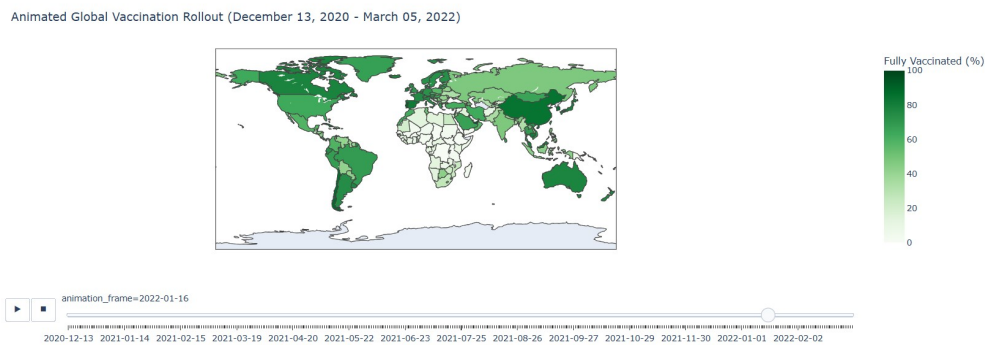


Figure 10: Global Vaccination Rollout Animation

4 Conclusions

This case study performed a rigorous, tri-faceted analysis of the OWID COVID-19 dataset, successfully meeting all project objectives. Our methodology—which emphasized structural filtering, justified NaN handling, and a two-dataframe approach—provided a robust foundation for analysis.

4.1 Synthesis of Findings

Our analysis generated three key conclusions:

1. Data-Driven Methodology is Critical: Our preliminary finding of systemic reporting lags (especially on weekends) proved that using 7-day smoothed data was not just a preference but a methodological necessity.

2. Demographics and Wealth Defined Outcomes: Demographics (median age) was the key predictor of mortality, while wealth (GDP) was the key predictor of vaccine access.

3. Vaccination Was the Decisive Intervention: We provided clear, visual evidence that the mass vaccination campaign successfully “decoupled” cases from deaths.

4.2 Limitations and Future Work

Limitations: This analysis is limited by the data's sparse nature for key metrics like `icu_patients`, `total_tests`, and `hosp_patients`. `Stringency_index` is a composite score that may not capture the nuance of specific policies.

Future Work: A future analysis could focus on `excess_mortality` (a more accurate measure of death) and integrate variants data to more precisely attribute waves of infection.

References

- [1] H. Ritchie, E. Mathieu, L. Rod  s-Guirao, C. Appel, C. Giattino, E. Ortiz-Ospina, J. Hasell, B. Macdonald, D. Beltekian, and M. Roser, "Coronavirus Pandemic (COVID-19)," *Our World in Data*, 2020. [Online]. Available: <https://ourworldindata.org/coronavirus>. [Accessed: Nov. 23, 2025].
- [2] World Health Organization, "WHO Coronavirus (COVID-19) Dashboard," 2023. [Online]. Available: <https://covid19.who.int/>. [Accessed: Nov. 23, 2025].